

How to Run the Notebook

1. **Download the dataset** from Kaggle and extract it locally.
Make sure the folder contains the quarterly folders such as 2019-q1, 2019-q2, up to 2021-q4.
2. **Open the notebook** in Jupyter Notebook or VSCode.
3. **Update the dataset path** in the notebook:

None

```
DATA_ROOT =  
r"your/local/path/to/ookla-internet-speed-dataset/versions/9"
```

4. **Install required libraries** if not yet installed:

None

```
pip install pyspark pandas numpy matplotlib
```

5. **Run the notebook cells in order.**
Do not skip cells because later sections depend on earlier DataFrames such as `project_df`, `featured_df`, `ml_df`, and `clustered_df`.
6. **Avoid unnecessary full-data actions** such as `.toPandas()`, `.cache()`, or `.count()` on the full dataset unless already specified in the notebook.
7. **Expect long runtimes.**
K-Means and regression training may take several minutes to over an hour depending on the machine.
8. **Recommended setup:**
Keep the laptop plugged in, disable sleep mode, and ensure enough free disk space for Spark temporary files.